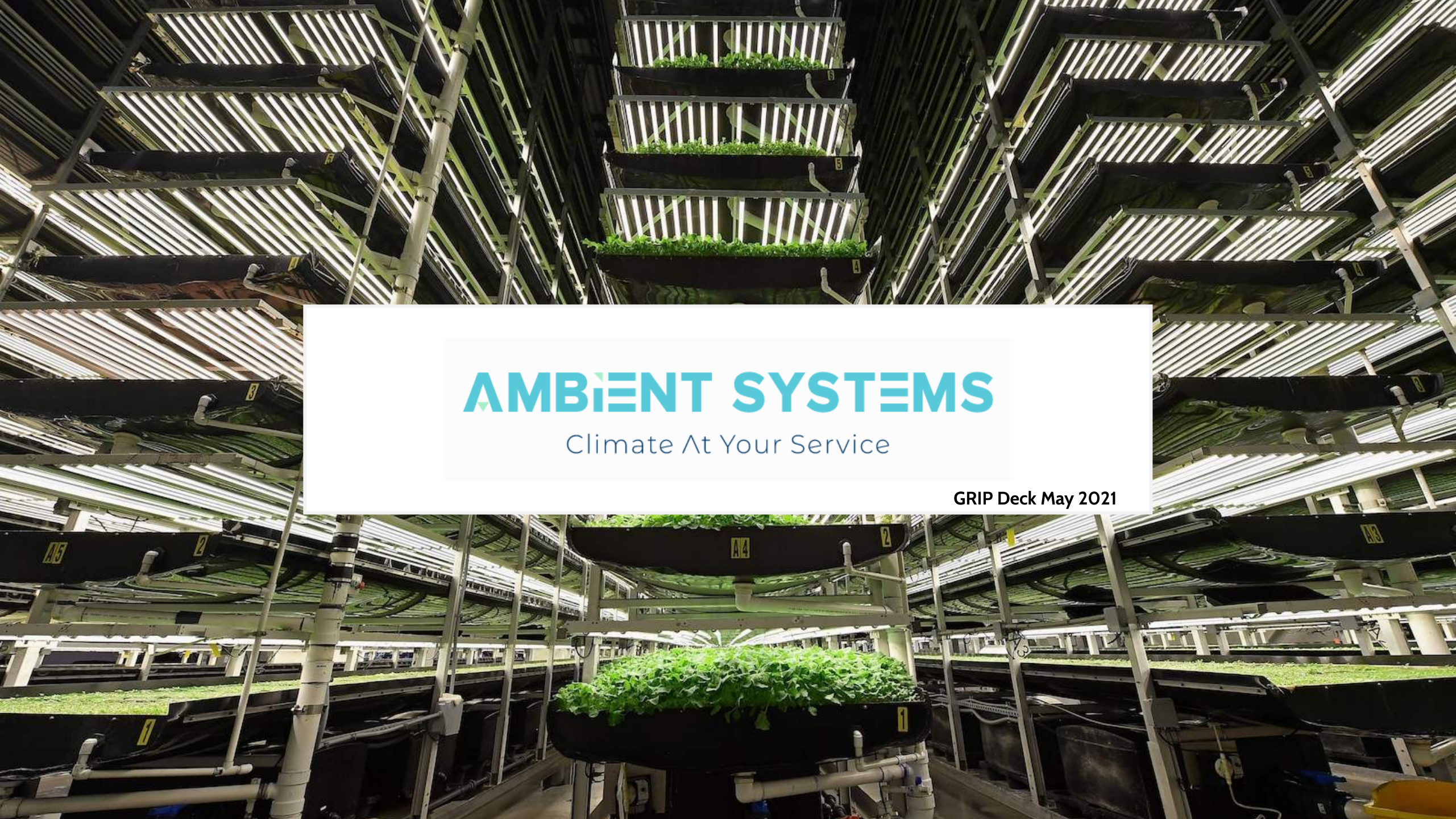


AMBIENT SYSTEMS

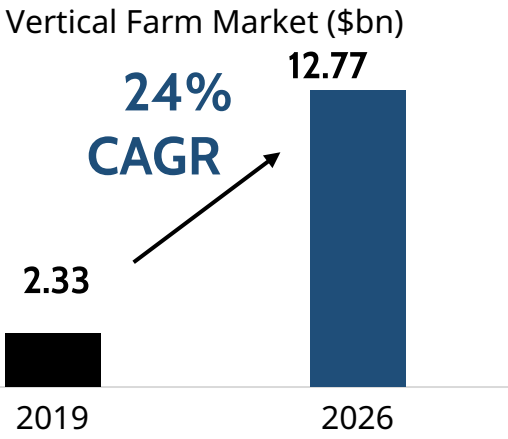
Climate At Your Service

GRIP Deck May 2021

Next Wave of APAC Indoor Farms at Double-Digit Growth



USD1.2BN
opportunity in vertical farms in Asia Pacific

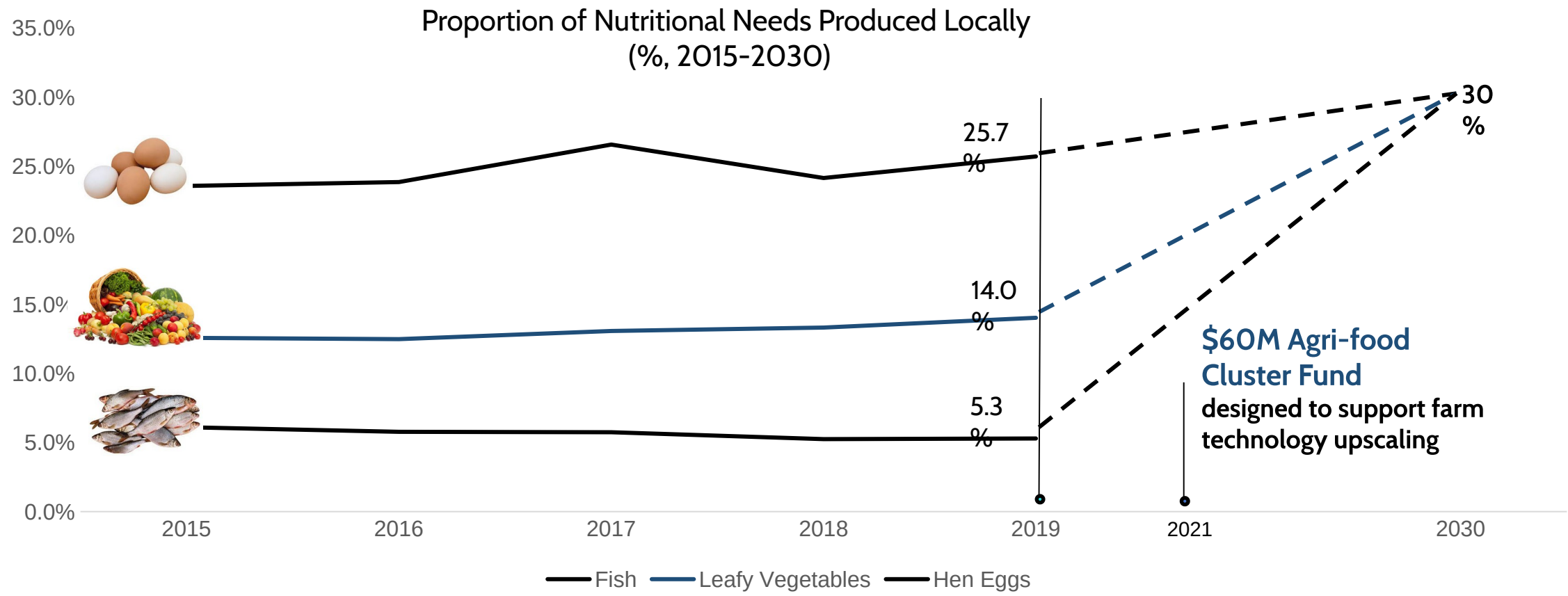


Source: Frost & Sullivan, McKinsey, Data Bridge

© Ambient Systems

AMBIENT SYSTEMS
Climate At Your Service

“30 by 30” Government Targets Raises Local Production Agenda

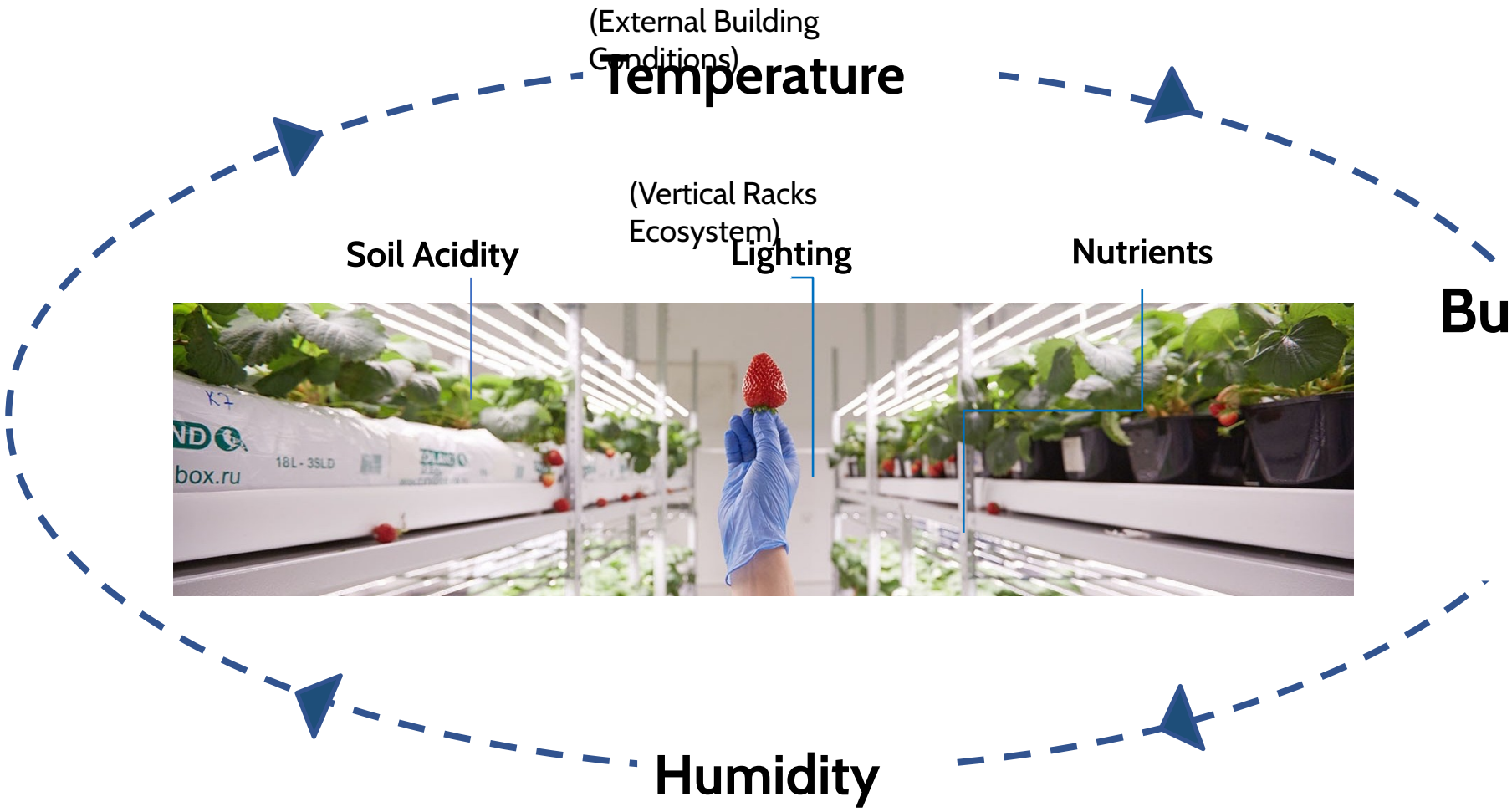


Source: Agri-food and Veterinary Authority, Singapore Food Agency, National Parks Board

© Ambient Systems

AMBIENT SYSTEMS
Climate At Your Service

Undesired Interactions Between Building Conditions and Crops

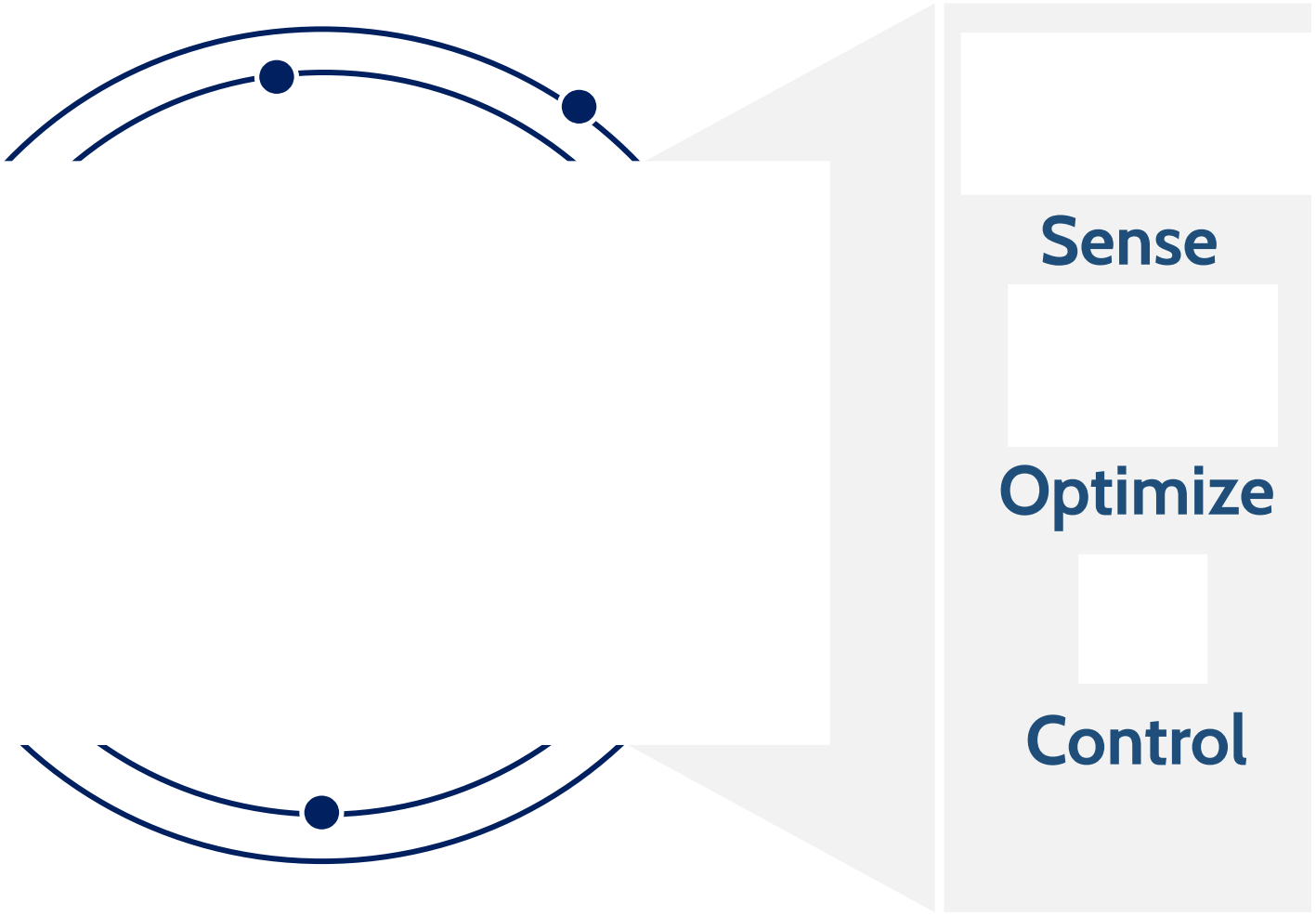


Building Conditions affect crops

- Inconsistent Environment
- Habitat for Mold and Pest
- Affect level of photosynthesis

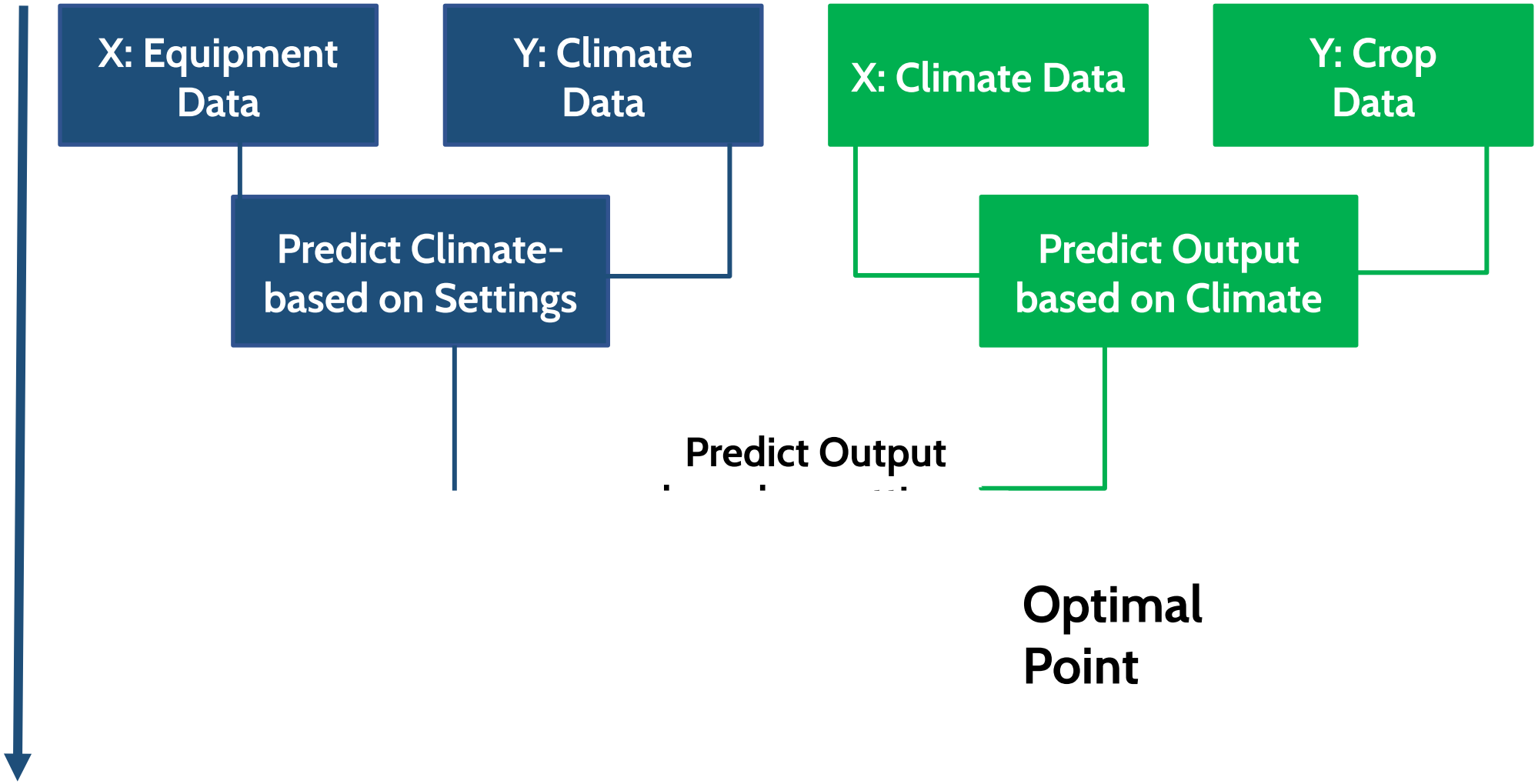
AMBIENT SYSTEMS
Climate At Your Service

Climate Management Platform for Indoor Farms



AMBIENT SYSTEMS
Climate At Your Service

Patent: Computer System for Management of Buildings with Indoor Farms



Singapore patent pending: 10202105064T

Three Stage Supervised Learning

Optimal Point

AMBIENT SYSTEMS
Climate At Your Service

Singapore patent pending: 10202105064T

Three Stage Supervised Learning

X: Equipment Data

Y: Climate Data

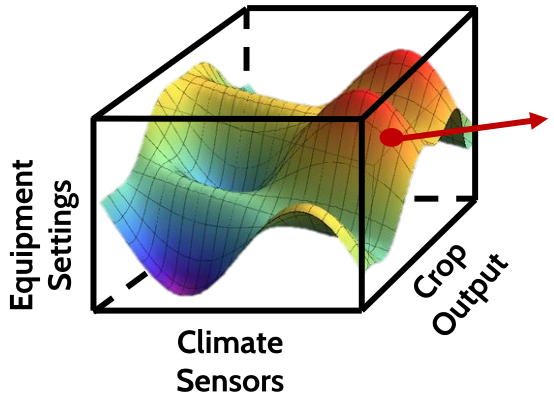
Predict Climate-based on Settings

X: Climate Data

Y: Crop Data

Predict Output based on Climate

Predict Output based on settings



Optimal Point

AMBIENT SYSTEMS
Climate At Your Service

Trade Secret: Operational Data on Creating an Optimal Indoor Climate

Climate Data

	Parameter
Current POC	Air Temperature
	Relative Humidity
	CO2
	Spectrum Light

Crop Data

	Parameter
Full-scale Plot	Soil Temperature
	Soil Moisture
	pH level
	Electrical Conductivity (EC_w)
	Crop specific parameters

Building Data

Equipment Settings
• Equipment setpoints • Light schedules • Chiller Plant settings • AHU Fan settings • Auxiliary Devices

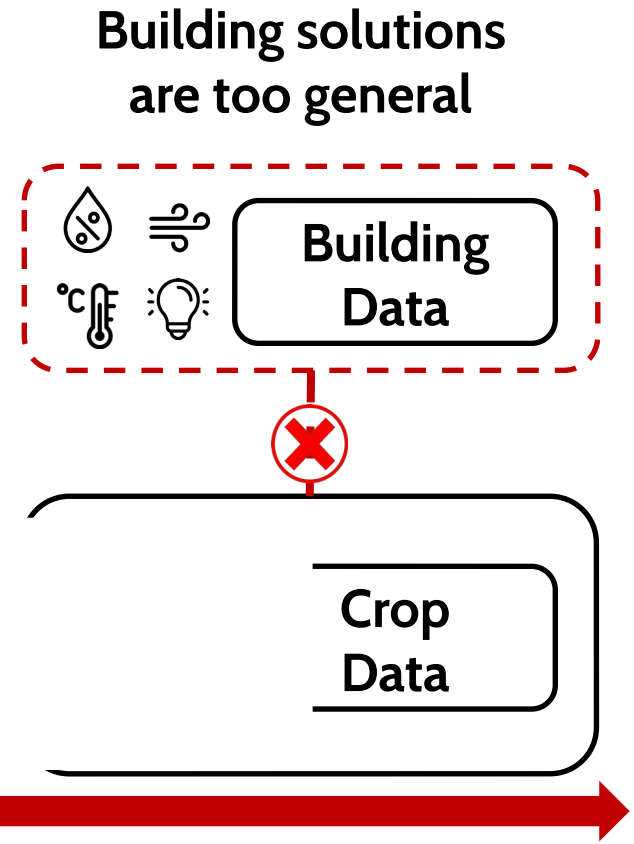
Proof of Concept: Strawberry Farm

Pilot Test	Findings	Potential Improvements
	CO2 at 25% of the optimal	• Increased photosynthesis by up to 75% • Earlier flowering • Higher fruit yields by up to 30%
	Temp at 70% of the optimal	• Increased leaf and petiole growth • Greater sucrose • More biomass at the roots

Ambient proprietary method provides equipment setpoints to achieve optimal targets

Time	Thermostat 1	Thermostat 2	Fan 1	Fan 2
May 28 – 2 pm	14 C	12 C	2 ACH	5 ACH

Indoor Farms Fail To Get Climate Management Right



© Ambient Systems

AMBIENT SYSTEMS
Climate At Your Service

We provide value beyond building controls and energy savings

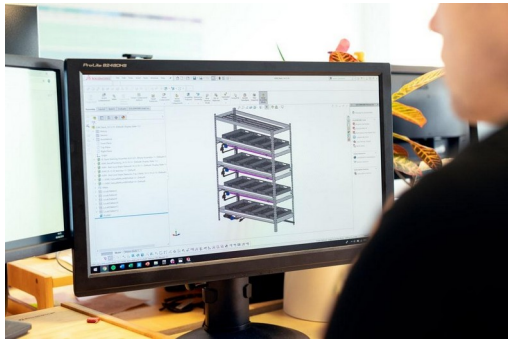


Source: Honeywell

Get Access Through A B2B Subscription

Yearly Subscription Model

Basic Monitoring



- Real-time data
- Online dashboards
- Push Alerts

Coming Soon

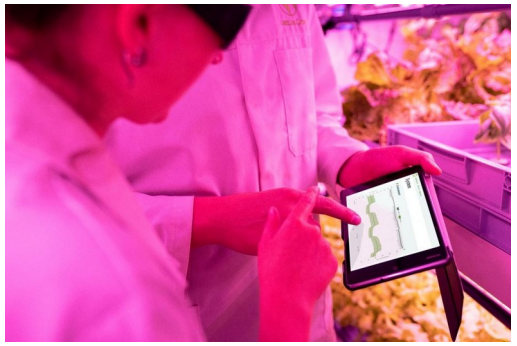
Automated Building Controls



- Full HVAC connectivity
- Real-time environmental adjustments

Coming Soon

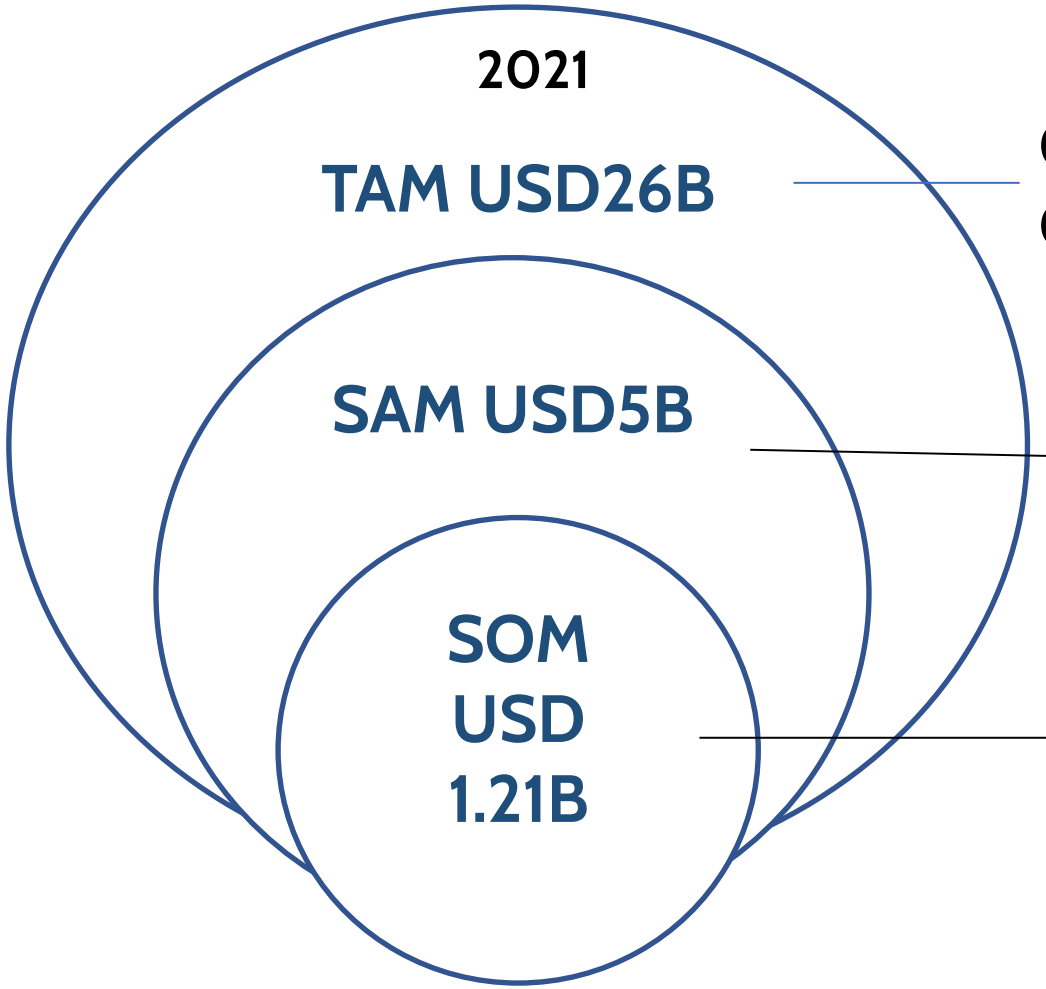
Advanced Insights



- Integrate user data
- Obtain Actionable Insights according to needs

One-time	**Installing Sensors** SGDXX / Sq ft	**Integration Cost** SGDXX / Sq ft	
Per Month	**Software Subscription** SGDXX / month	**Cost of Maintenance** SGDXX / month	**Additional Insights** SGDXX / month

Market Opportunity: TAM, SAM, SOM



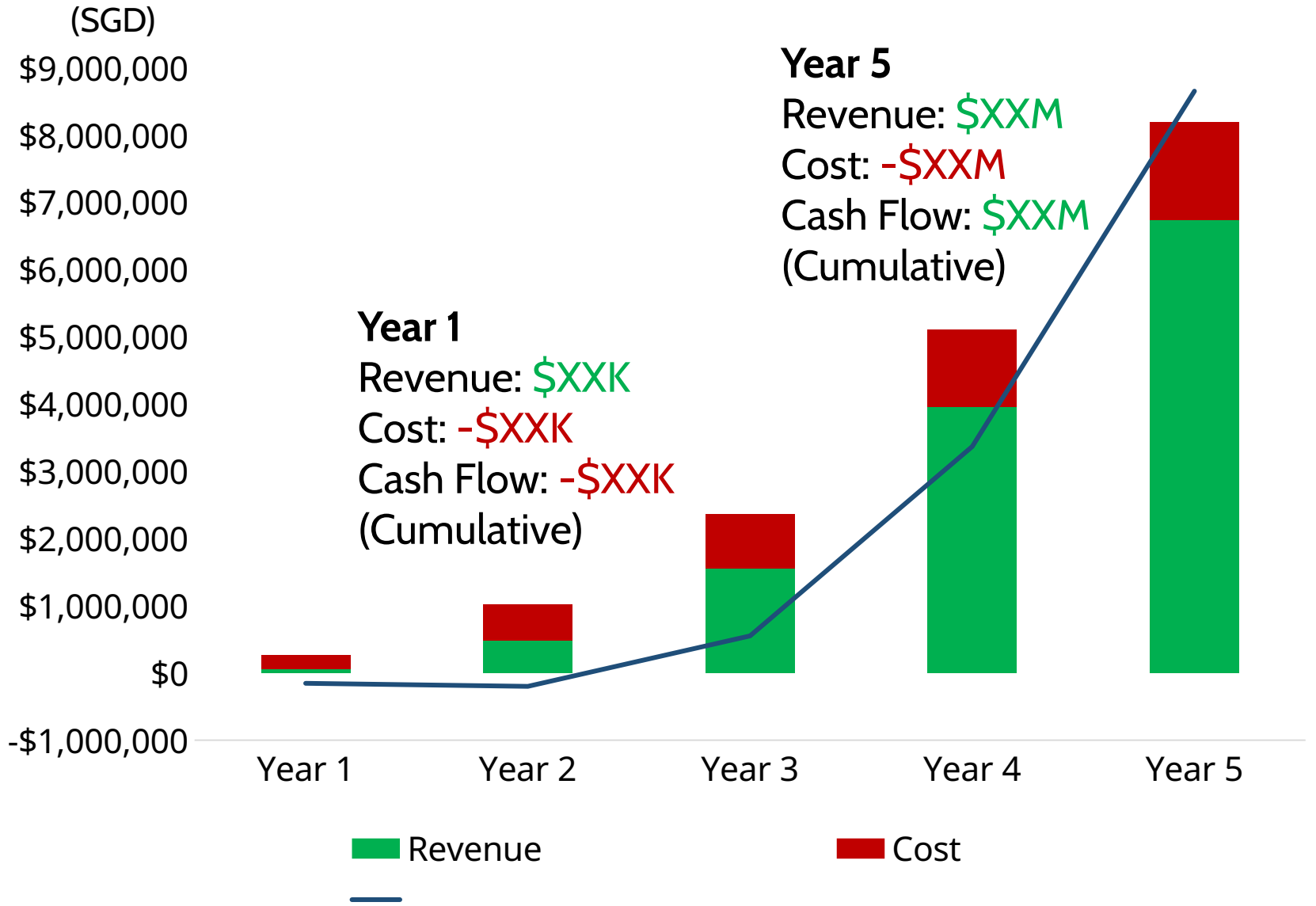
Global HVAC Automation and Controls

AI in Building Automation

APAC
Vertical Farming

Source: Frost & Sullivan, McKinsey, Data Bridge

Our financial projections



XX%
Average Gross Profit

$XXM
NPV

AMBIENT SYSTEMS
Climate At Your Service

We have a funding gap of XX over the next XX months

July 2021

Dec 2022

Market Expansion

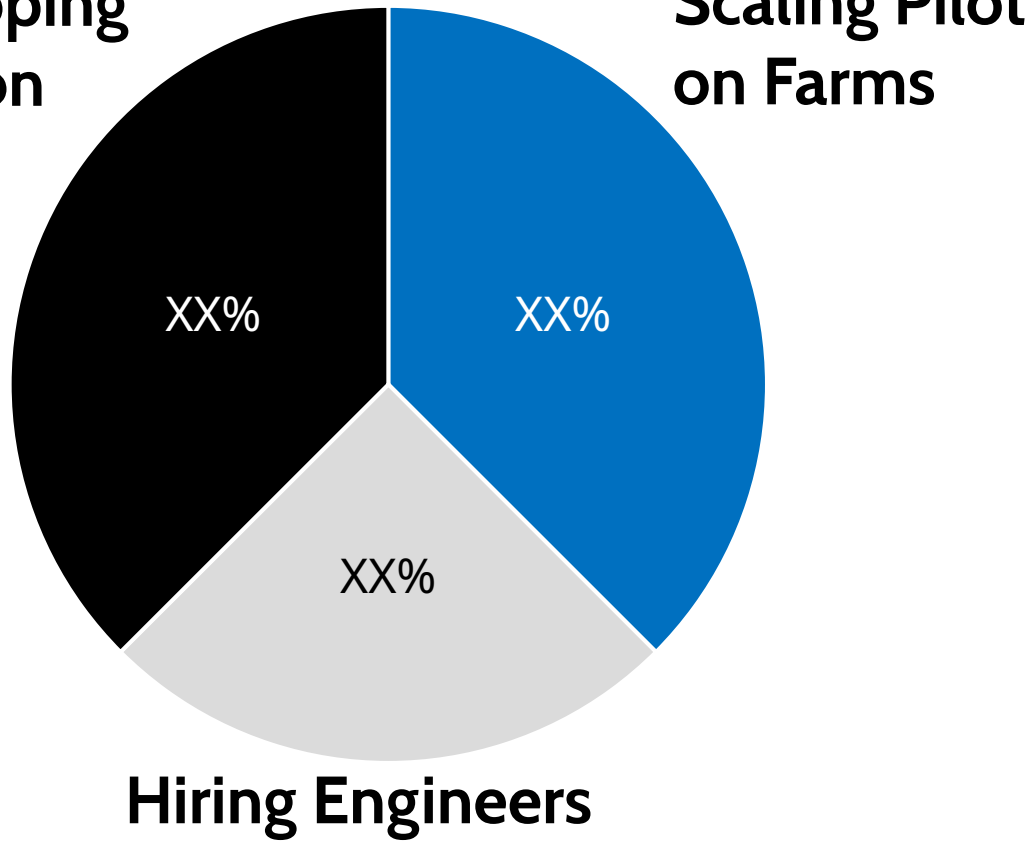


Offerings

- Monitoring
- User Insights

Segments

- Farms



AMBIENT SYSTEMS
Climate At Your Service

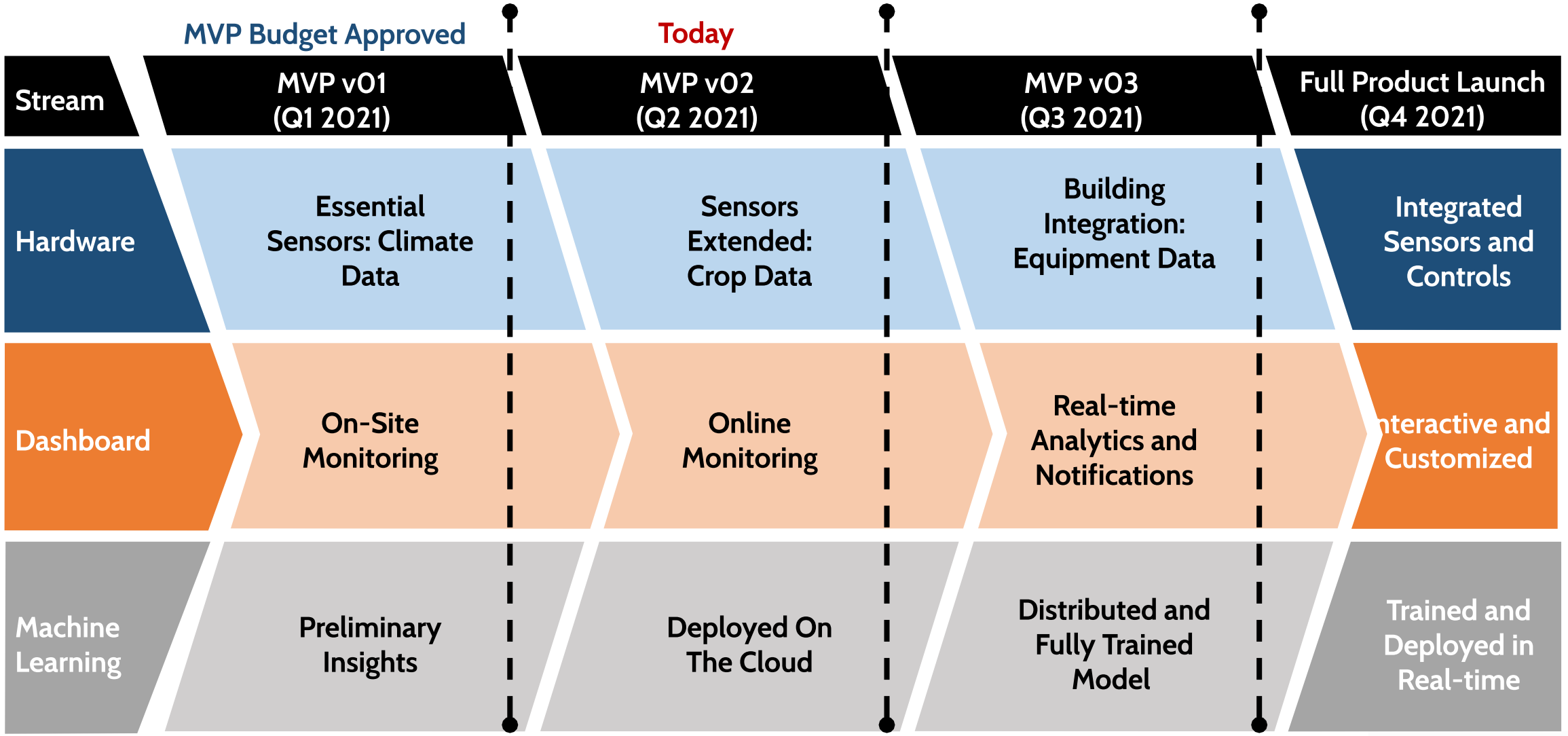
Product Validation from Farms



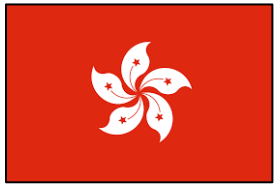
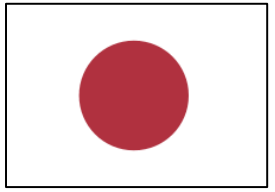
We have received X Letter of Interests worth up to

SGDXXK

Full product launch on-track by end of the year



Aim to offer full-suite solution by Japan expansion

	2021	2022	2023	2024	2025
Market Expansion					
Offerings	Monitoring User Insights		Control		
Segments	Farms		Healthcare		Restaurants

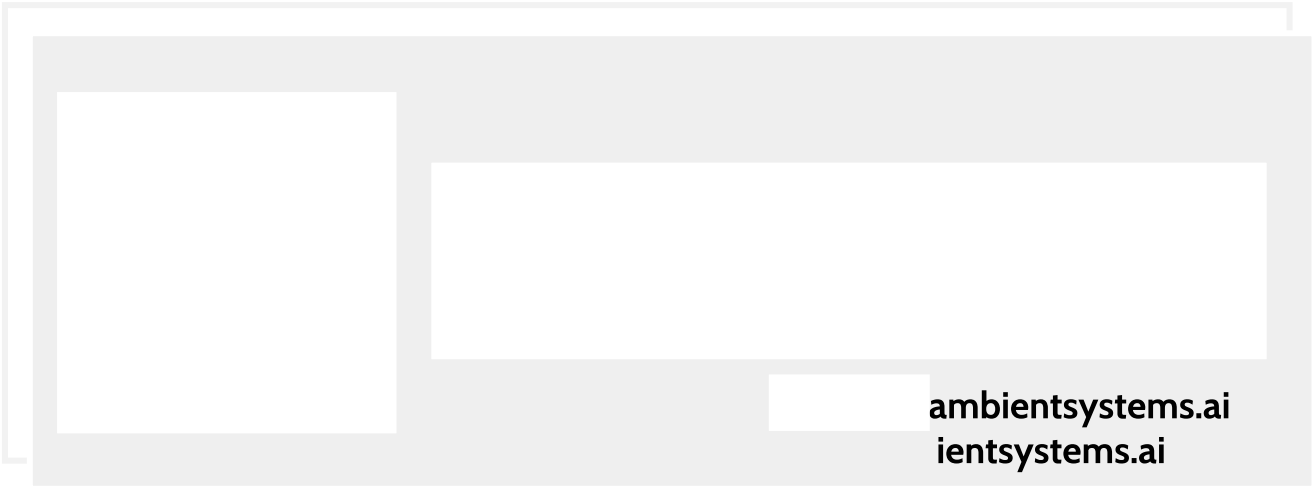
AMBIENT SYSTEMS
Climate At Your Service

About Us

- **MComp, Computer Science**
  National University of Singapore
- **BASc, Civil Engineering, Sustainable Energy**
  University of Toronto
- **Serial Entrepreneur**
  5 years experience in cleantech, deep tech

, Finance
Singapore Management University

- **Strategy and Fundraising**
  3 years experience across
  e-commerce, gaming and payments



1. Experienced Team with tech, strategy and fundraising background

2. Our Indoor Farming launch market growing at a CAGR of 24%

3. Received with Indoor Farms

4. Full Product Launch by End of 2021

5. Clear Go-To-Market Strategy towards Agri-tech Hubs